

Education

CHONGQING UNIVERSITY OF TECHNOLOGY, CHINA

2015-2019

Bachelor of Technology, Software Engineering

- **Cumulative Grade Point Average** - 8.63/10.00; Ranking in 5% of the grade (6/150).
- **Coursework** - Data Structures and Algorithms, Computer Networking, Data Mining, Operating System.
- First-class Scholarship at school Level (Top 5%).
- Excellent Bachelor's Degree Thesis at School Level (Top 3%).

Work Experience

RESEARCH INTERN – Big Data Research Center, Duke Kunshan University, Suzhou

Feb 2019 – Jun 2019

- Responsible for manufacturing data analysis of silicon wafer production plant.
- Responsible for product defect detection device and computer vision algorithm development.
- Campus reception and technical demonstration.

Research Interests

- Machine Learning, Computer Vision
- Computer Vision Based Weather Analysis and Application, Smart Meteorology
- Intrinsic Image Decomposition, Transfer Learning

Latest Projects

Generating the Cloud Motion Winds Field from Satellite Cloud Imagery Using Deep Learning Approach. [\[URL\]](#)

- **Author:** Chao Tan (Independent author).
- **Description:** First proposed to estimate cloud motion winds field using deep network, and open source a large training dataset.
- **Publication:** IEEE International Geoscience and Remote Sensing Symposium (IGARSS). IEEE, 2021.

Unsupervised Shadow Removal Using Target Consistency Generative Adversarial Network. [\[URL\]](#)

- **Authors:** Chao Tan (First author), Xin Feng.
- **Description:** Implemented a novel algorithm to remove natural scene shadows in an unsupervised manner efficiently.
- **Publication:** Submitted.

TCLNet: Learning to Locate Typhoon Center Using Deep Neural Network. [\[URL\]](#)

- **Author:** Chao Tan (Independent author).
- **Description:** Proposed a large-scale dataset and lightweight deep neural network to automatically locate the cyclone center.
- **Publication:** IEEE International Geoscience and Remote Sensing Symposium (IGARSS). IEEE, 2021.

Improving Typhoon Center Location Models by Augmented Typhoon Image and Distillation Methods. [\[URL\]](#)

- **Author:** Chao Tan (Independent author).
- **Description:** Used unconditional GAN and knowledge distillation strategy to achieve SOTA typhoon center location performance.

FORECAST-CLSTM: A New Convolutional LSTM Network for Cloudage Nowcasting. [\[URL\]](#)

- **Authors:** Chao Tan (First author), Xin Feng, Jianwu Long and Geng Li.
- **Description:** Proposed a hierarchical convolutional LSTM network, a novel loss function and a new dataset for cloudage nowcasting.
- **Publication:** IEEE Visual Communications and Image Processing (VCIP). IEEE, 2018 (Oral presentation).

Skills & Qualifications

Programming Languages/Frameworks: Python, Java, C, PyTorch, TensorFlow, Keras, JavaScript, Matlab
Technologies: MySQL Server, Maple, Android Development, Minitab
Qualifications: Deep Learning Associate (Baidu Technology Certification).